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Quiz for Week 5

Graded Quiz • 20 min

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Assessments

Quiz: Quiz for Week 5

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Quiz for Week 5

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According to the instructor, what are the major sources of costs in the facility location problem introduced in this week? Check all the correct answers.

☒ Office rent.

☒ Correct

☒ Travelling cost.

☒ Correct

☒ Engineer hosting cost.

☒ Correct

☐ Engineer training cost.

☐ Facility construction cost.

1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

2. Which of the following statements best describes the main decision to make in the facility location problem discussed in this week?

1 / 1 point

☐ For each location, determine the facility scale and facility type.

☒ For each location, determine the facility scale and number of engineers hosted here.

☐ For each location, determine the number of engineers hosted here.

☐ For each location, determine whether to build a facility there.

☐ None of the above.

☒ Correct

3. What of the following is not one of the three models we need in an Operations Research study?

1 / 1 point

☐ A conceptual model.

☐ A mathematical model.

☒ A universal model.

☐ A computer model.

☐ All of the above.

☒ Correct

4. Recall that in the facility location problem discussed in this week, there is a constraint saying that for each customer and each location, only a built facility may serve customers. The mathematical formulation is

1 / 1 point

$$y_{ij} \leq \sum_{k \in K} x_{jk} \quad \forall i \in I, j \in J.$$

Suppose that the formulation is changed to

$$y_{ij} \leq x_{jk} \quad \forall i \in I, j \in J, k \in K.$$

Which of the following statement is correct?

☒ The new feasible region becomes smaller or the same. The new formulation is wrong.

☐ The new feasible region remains the same. The new formulation is correct.

☐ The new feasible region becomes larger or the same. The new formulation is wrong.

☐ The new feasible region may become smaller or larger. The new formulation is wrong.

☐ None of the above.

☒ Correct

5. Which of the following is the most appropriate category of the heuristic algorithm introduced in this week?

1 / 1 point

☐ A brute-force algorithm.

☐ A divide-and-conquer algorithm.

☒ A greedy algorithm.

☐ A dynamic programming algorithm.

☐ A backtracking algorithm.

☒ Correct

